Informatica® Cloud Data Integration

SAP OData V4 Connector

Informatica Cloud Data Integration SAP OData V4 Connector
October 2026

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Preface

Use SAP OData V4 Connector to learn how to read from or write to OData V4-compliant applications in SAP by using Data Integration. Learn to create an SAP OData V4 connection, develop and run mappings, and mapping tasks in Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to SAP OData V4 Connector

You can use SAP OData V4 Connector to access SAP applications from Data Integration. Use the SAP OData V4 connection to securely read data from or write data to entities that are exposed and accessible through OData V4-compliant applications in SAP.

SAP OData V4 Connector communicates with the SAP OData APIs version 4 that are exposed through the SAP OData V4 service and accesses SAP applications. The SAP OData V4 service helps you to process SAP data in a standardized way in the data integration process.

You can use SAP OData V4 Connector to perform the following operations:

- Read data from and write data to hierarchical and flat entities in SAP S/4HANA applications.
- Read flat data from graphical views and analytical models in SAP Datasphere applications.

When you use SAP OData V4 Connector, you can create an SAP OData V4 connection and use it both as a source and a target in a mapping, dynamic mapping task, and mapping task. When you run mappings, dynamic mapping tasks, or mapping tasks, the Secure Agent reads from and writes to the OData V4-compliant application in SAP based on the workflow and SAP OData V4 connection configuration.

You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality.

SAP OData V4 Connector assets

Create assets in Data Integration to integrate data using SAP OData V4 Connector.

When you use SAP OData V4 Connector, you can include the following Data Integration assets:

- Dynamic mapping task
- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for SAP OData V4

Create an SAP OData V4 connection to securely read from or write to OData V4-compliant applications, such as SAP Datasphere and SAP S/4HANA applications.

You can use an SAP OData V4 connection to specify sources and targets in mappings, dynamic mapping tasks, and mapping tasks.

Prepare for authentication

You can configure basic, authorization code, and client credentials authentication types to connect to OData V4-compliant applications in SAP.

Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

Basic

Use basic authentication to connect to SAP S/4HANA applications. You need the SAP account user name and password to configure basic authentication.

Get the required details from the SAP S/4HANA application to which you want to connect.

For more information about basic authentication in SAP, see Basic authentication in the SAP documentation.

Authorization code

Use OAuth 2.0 authorization code authentication to connect to SAP Datasphere and SAP S/4HANA applications. You need the SAP client ID, client secret, authorization token URL, access token URL, and access token to configure OAuth 2.0 authorization code authentication.

Connecting to SAP Datasphere applications

To connect to SAP Datasphere applications using OAuth 2.0 authorization code authentication, you need to complete certain prerequisite tasks.

Create an authorization integration in SAP, and register the following Informatica redirect URL in SAP Integration Suite:

`https://<Informatica cloud hosting facility for your organization>/ma/proxy/oauthcallback`

If the access token expires and the response returns a 401 error code, the Informatica redirect URL, which is outside the customer firewall, tries to connect to the endpoint and retrieves a new access token.

For more information about how to create an authorization integration and get the authorization details, see OAuth 2.0 authorization code in the SAP documentation.

Connecting to SAP S/4HANA applications

To connect to SAP S/4HANA applications using OAuth 2.0 authorization code authentication, you need to complete certain prerequisite tasks.

- Create an authorization integration in SAP, and register the following Informatica redirect URL in SAP Integration Suite:
  `https://<Informatica cloud hosting facility for your organization>/ma/proxy/oauthcallback`

  If the access token expires and the response returns a 401 error code, the Informatica redirect URL, which is outside the customer firewall, tries to connect to the endpoint and retrieves a new access token.

  For more information about how to create an authorization integration and get the authorization details, see OAuth 2.0 authorization code in the SAP documentation.
- Import the `.cer` certificate into the cacerts file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure Agent installation directory>/jdk/jre/lib/security`
  - `<Secure Agent installation directory>/jdk/lib/security`
  - `<Secure Agent installation directory>/jdk8/jre/lib/security`

  Ensure to add the certificate to all the available directories.
- If there is a jdk directory within the `<Secure Agent installation directory>\apps` folder, navigate to the following directory and import the `.cer` certificate into the `cacerts` file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure Agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`
  - `<Secure Agent installation directory>\apps\jdk\zulu17<latest_version>\lib\security`
- Restart the Secure Agent after you import the `.cer` certificate into the cacerts file.

Client credentials

Use OAuth 2.0 client credentials authentication to connect to SAP S/4HANA applications. You need the SAP client ID, client secret, access token URL, scope, and access token to configure OAuth 2.0 client credentials authentication.

To use OAuth 2.0 client credentials authentication, you need to complete certain prerequisite tasks.

- Configure the OAuth endpoint with the client credentials grant type, and then create an authorization integration to get the authorization details in the SAP system.
  For more information about how to create an authorization integration and get the authorization details, see OAuth 2.0 client credentials in the SAP documentation.
- Import the `.cer` certificate into the cacerts file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure Agent installation directory>/jdk/jre/lib/security`
  - `<Secure Agent installation directory>/jdk/lib/security`
  - `<Secure Agent installation directory>/jdk8/jre/lib/security`

  Ensure to add the certificate to all the available directories.

- If there is a jdk directory within the `<Secure Agent installation directory>\apps` folder, navigate to the following directory and import the `.cer` certificate into the `cacerts` file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure Agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`

  - `<Secure Agent installation directory>\apps\jdk\zulu17<latest_version>\lib\security`
- Restart the Secure Agent after you import the `.cer` certificate into the cacerts file.

Connect to SAP OData V4

Let's configure the SAP OData V4 connection properties to connect to SAP Datasphere or SAP S/4HANA applications.

Before you begin

Before you get started, you'll need to get the relevant configuration details from your SAP account based on the authentication type that you want to configure.

Check out "Prepare for authentication" on page 7 to learn more about the authentication prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, serverless, or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Property	Description
SAP Application Type	The SAP application type of the SAP OData V4 service endpoint. Select one of the following application types from the list: - SAP Datasphere. Connects to an SAP Datasphere application. Configure the OAuth 2.0 authorization code authentication type to access SAP Datasphere. - SAP S/4Hana. Connects to an SAP S/4HANA application. Configure the basic, OAuth 2.0 authorization code, or OAuth 2.0 client credentials authentication type to access SAP S/4HANA. Default is SAP Datasphere.
Service Type	The service type of the SAP OData V4 service endpoint to which you want to connect. Select one of the following service types from the list: - Default. Connects to a specific SAP service. - SAP Gateway Catalog. Connects to the SAP catalog service that provides a list of all available services on SAP Gateway. Default is Default.

Service URL	The service URL for the selected application type and service type.
	Enter the service URL in the following format depending on the selected application type and service type:
	- For the **SAP Datasphere** application type and the **Default** service type:
	`https://<Host name of the SAP server>/api/v1/dwc/consumption/relational/<Space name of SAP Datasphere>/<Asset name of SAP Datasphere>/`
	- For the **SAP Datasphere** application type and the **SAP Gateway Catalog** service type:
	`https://<Host name of the SAP server>/api/v1/dwc/catalog/`
	- For the **SAP S/4Hana** application type and the **Default** service type:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata4/sap/<Service name>/<Path of the service>/<version number>/`
	- For the **SAP S/4Hana** application type and the **SAP Gateway Catalog** service type:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata4/iwfnd/config/default/iwfnd/<Catalog service name in SAP S/4HANA>/<version number>/`
	Additionally, you can append multiple SAP query parameters to the service URL to connect to SAP Datasphere or SAP S/4HANA applications, separated by an ampersand (&), in the following format:
	`<Custom query1>=<value>&<Custom query2>=<value>&<Custom query3>=<value>....`
	When you add SAP query parameters, ensure that there is no space before and after the equal sign (=).
	For more information about the list of SAP query parameters that you can configure, see SAP URL parameters in the SAP documentation.

Authentication types

You can configure the basic, authorization code, and client credentials authentication types to access OData V4-compliant applications, such as SAP Datasphere and SAP S/4HANA applications.

Select the required authentication type and then configure the authentication-specific parameters.

Basic authentication

The minimum requirement for a basic authentication is the SAP account user name and password. Use the basic authentication type to connect to SAP S/4HANA applications.

The following table describes the connection properties for basic authentication:

Property	Description
User Name	The user name to connect to the SAP S/4HANA application.
Password	The password for the user name to connect to the SAP S/4HANA application.

Authorization code authentication

OAuth 2.0 authorization code authentication is the default type and the minimum requirement for an OAuth 2.0 authorization code authentication is the SAP client ID, client secret, authorization token URL, access token URL, and access token.

Use the OAuth 2.0 authorization code authentication type to connect to SAP Datasphere and SAP S/4HANA applications.

The following table describes the basic connection properties for OAuth 2.0 authorization code authentication:

Property	Description

Authorization Token URL	The SAP OAuth 2.0 authorization server that is used to authorize the user request.
	Enter the authorization token URL in the following format:
	`https://<Host name of the server>/oauth/authorize`
	For example, if the host name is example.authentication.us01.hana.ondemand.com, enter the following authorization token URL:
	`https://example.authentication.us01.hana.ondemand.com/oauth/authorize`

Access Token URL	The SAP OData V4 access token endpoint that is used to exchange the authorization code for an access token.
	Enter the access token URL in the following format:
	`https://<Host name of the server>/oauth/token`
	For example, if the host name is example.authentication.us01.hana.ondemand.com, enter the following access token URL:
	`https://example.authentication.us01.hana.ondemand.com/oauth/token`

Client ID	The client identifier of your application generated when you configure the application for OAuth.

Property	Description
Client Secret	The client secret generated for the client ID.
Access Token	The access token granted by the authorization server to access the SAP data. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties for OAuth 2.0 authorization code authentication:

Property	Description
Scope	The scope of the access request when the SAP OData V4 endpoint has defined custom scopes. You can enter multiple scope attributes, each separated by a space, in the following format: `<Scope attribute1> <Scope attribute2> <Scope attribute3>....`

Client credentials authentication

The minimum requirement for an OAuth 2.0 client credentials authentication is the SAP client ID, client secret, access token URL, scope, and access token.

Use the OAuth 2.0 client credentials authentication type to connect to SAP S/4HANA applications.

The following table describes the basic connection properties for OAuth 2.0 client credentials authentication:

Property	Description

Access Token URL	The SAP OData V4 access token endpoint that is used to exchange the authorization code for an access token.
	Enter the access token URL in the following format:
	`https://<Host name of the server>:<Port number>/sap/bc/sec/oauth2/token?sap-client=<SAP client number>`
	For example, if the host name is abcd01con02.example.com and port number is 44301, enter the following access token URL when the SAP client number is 800:
	`https://abcd01con02.example.com:44301/sap/bc/sec/oauth2/token?sap-client=800`

Client ID	The client identifier of your application generated when you configure the application for OAuth.
Client Secret	The client secret generated for the client ID.

Property	Description
Scope	The scope of the access request when the SAP OData V4 endpoint has defined custom scopes. You can enter multiple scope attributes, each separated by a space, in the following format: `<Scope attribute1> <Scope attribute2> <Scope attribute3>....`
Access Token	The access token granted by the authorization server to access the SAP data. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties for OAuth 2.0 client credentials authentication:

Property	Description
Access Token Parameters	Additional parameters to use with the access token URL. Define the access token parameters in the following JSON format: `[{"Name":"<Parameter name>","Value":"<Parameter value>"}]` For more information about the access token parameters that you can define, see the SAP documentation.
Client Authentication	The method of sending client authentication details for authorization when you connect to the SAP OData V4 service. Select one of the following client authentications from the list: - Send client credentials in body. Sends the client ID and client secret for authorization in the body of the request. - Basic auth header. Sends the client ID and client secret for authorization in the header of the request. Default is **Send client credentials in body**.

Configure a load balancing connection

When you use the basic authentication type to connect to the SAP S/4HANA server, you can create a load balancing connection. It allows you to connect to the SAP S/4HANA server with the least load at run time.

To configure a load balancing connection, you need to perform the prerequisite tasks.

1. Import the `.cer` certificate into the cacerts file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:
   - `<Secure Agent installation directory>/jdk/jre/lib/security`

- `<Secure Agent installation directory>/jdk/lib/security`
- `<Secure Agent installation directory>/jdk8/jre/lib/security`

Note: Ensure that you import the certificate to all the available directories.

2. If a jdk directory is available within the `<Secure agent installation directory>\apps` folder, navigate to the following directory and import the `.cer` certificate into the `cacerts` file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:
   - `<Secure Agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`
   - `<Secure Agent installation directory>\apps\jdk\zulu17<latest_version>\lib\security`
3. Restart the Secure Agent.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use an unauthenticated proxy server on Windows and Linux.

The proxy settings apply to connections used in mappings and in mappings in advanced mode.

To configure the proxy settings for the Secure Agent, configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.

Bypass the proxy server

You can bypass the proxy server for specific hosts, such as internal or trusted hosts.

When you route Secure Agent traffic through a proxy server, some hosts might require a direct connection. Configure the proxy.ini file so that the Secure Agent connects to those hosts directly instead of through the proxy server.

The proxy.ini file stores the proxy server configuration that the Secure Agent uses for outbound connections. Use the `InfaAgent.NonProxyHost` property to identify the hosts that the Secure Agent connects to directly instead of through the proxy server.

1. Go to the following directory on the Secure Agent machine:

   `<Secure Agent installation directory>/apps/agentcore/conf`
2. Open the proxy.ini file and enter the hosts to bypass in the `InfaAgent.NonProxyHost` property.

   For example, to bypass the proxy server for a wildcard domain, localhost, an IP address, and a specific host, enter a value in the following format:

   ```
   InfaAgent.NonProxyHost=*.<domain>|localhost|<IP address>|<host name>
   ```

   Note: You can enter multiple hosts, each separated by a pipe character.
3. Save and close the proxy.ini file.
4. Restart the Secure Agent.
5. Test the SAP OData V4 connection to verify that the Secure Agent connects to the specified hosts directly, without routing through the proxy server.

CHAPTER 3

Mappings and mapping tasks with SAP OData V4 Connector

You can configure a mapping to read from or write to OData V4-compliant applications, such as SAP Datasphere and SAP S/4HANA applications.

You can configure a Source or Target transformation to represent an SAP OData V4 object. Add the Source and Target transformations in the mapping canvas and configure the SAP OData V4 source and target properties accordingly. Use the Data Integration Mapping Designer to define and configure a mapping for SAP OData V4 sources and targets.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

Before you begin

Before you configure and run a mapping, verify that you have the required user permissions on SAP OData V4 objects that you read from SAP Datasphere applications.

The SAP administrator needs to configure the read only permission for the following roles in SAP to read from SAP Datasphere applications in SAP OData V4 mappings:

Catalog Administrator	DW Scoped Consumer	DW Viewer
Catalog User	DW Scoped Integrator	ZDW_Consumer
DW Administrator	DW Scoped Viewer	ZDW_Intergrator
DW Consumer	DW Scoped Space Administrator	ZDW_Space_admin
DW Integrator	DW Space Administrator	-
DW Modeler	DW Support User	-

For more information about the user permissions to read from SAP Datasphere, see Privileges and permissions.

SAP OData V4 sources in mappings

To read data from an SAP OData V4 source in the SAP Datasphere or SAP S/4HANA application, configure an SAP OData V4 object as a source in a mapping.

Specify the name and description of the SAP OData V4 source. Configure the source, query options, and advanced properties for the source object.

The following table describes the source properties that you can configure for an SAP OData V4 source:

Property	Description
Connection	Name of the source connection. You can select an existing connection or create a new connection for the source connection property.
Object Type	Type of the source object. Select **Single Object** as the object type. Default is Single Object.
Object	Name of the SAP OData V4 source object based on the object type selected. Select the source object for a single source. You can also select an existing object from the list.

The following table describes the query options that you can configure for an SAP OData V4 source:

Property	Description
Filter	Filters record based on the filter condition in a read operation. You can configure a simple filter or an advanced filter to filter rows from the source based on a filter condition. You can improve efficiency by filtering early in the data flow. Click **Configure** to add conditions to filter records from the data that the Secure Agent reads from the source. Select one of the following filter options from the list: - Not parameterized. Uses a basic filter to specify the object, field, operator, and value to select specific records. **Note:** Don't use the IS NULL or IS NOT NULL operators in a simple filter. - Advanced. Uses an advanced filter to define a complex filter condition. For more information on filters and operators that you can configure, see URI conventions for OData Version 2.0.
Sort	Add conditions to sort records. Select one of the following sort options from the list: - Not parameterized. Select the fields and type of sorting to use. - Parameterized. Use a parameter to specify the sort option.

The following table describes the advanced source properties that you can configure for an SAP OData V4 source:

Property	Description
Page Size	Determines the number of rows to read in each request. Default is 1000.
Number of Rows to Skip	Sets the number of rows you can skip. For example, you can set this property to n to skip the first n rows in a source and read from the (n+1) row. Default is 0.
Number of Rows to Fetch	Determines the number of rows you can read from the source. For example, specify a value of 100 to read 100 records from the source. Specify 0 to read all records from the source. Default is 0.
Data Serialization Format	The format of data that is transferred over the network. Select one of the following formats from the list: - ATOM/XML - JSON Default is JSON.
Tracing Level	Determines the amount of detail that appears in the log file. Select one of the following options from the list: - Terse - Normal - Verbose Initialization - Verbose Data Default is Normal.
Cache Size for Web Service Response (KB)	The memory in KB available for the web service response. If the web service response contains many rows or columns, you might want to increase the cache size. Default is 1024.
Optimize transformation	Select to improve performance for processing records at run time. Default is disabled.

SAP OData V4 targets in mappings

To write data to an SAP OData V4 target in the SAP S/4HANA application, configure an SAP OData V4 object as a target in a mapping.

Specify the name and description of the SAP OData V4 target. Configure the target and advanced properties for the target object.

The following table describes the target properties that you can configure for an SAP OData V4 target:

Property	Description
Connection	Name of the target connection. You can select an existing connection or create a new connection for the target connection property.
Object Type	Type of the target object. Select **Single Object** as the object type. Default is Single Object.
Object	The target object for the task. Select the target object for a single target.
Operation	Type of the target operation. Select one of the following operations from the list: - Insert - Update - Upsert (Update or Insert) - Delete - Data Driven. Not applicable Default is Insert.
Update Columns	This property appears only if you select update, upsert, or delete as the operation type. Select the unique key column as the condition field for the update, upsert, or delete operation in an SAP OData V4 target.
Data Driven Condition	Not applicable.

The following table describes the advanced target properties that you can configure for an SAP OData V4 target:

Property	Description
Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops them. By default, the mapping task forwards rejected rows to the next transformation. If you select the **Forward Rejected Rows** option, the agent flags the rows as rejected and writes them to the reject file. If you don't select the **Forward Rejected Rows** option, the Secure Agent drops the rejected rows and writes them to the session log file. The Secure Agent does not write the rejected rows to the reject file. Default is enabled.
Cache Size for Web Service Request (KB)	The memory in KB available for the web service request. If the target contains many rows or columns, you might want to increase the cache size. Default is 1024.
Transaction Commit Control	Not applicable.

Property	Description
Transformation Scope	The method in which the mapping applies the transformation logic to incoming data. Select one of the following transformation scopes from the list: - All Input. Processes data from a source that is not real-time. - Transaction. Processes data from a real-time source. Default is All Input. You can connect only one effective real-time source to a real-time target.
Optimize transformation	Select to improve performance for processing records at run time. Default is disabled.

Rules and guidelines for SAP OData V4 objects

Consider the following rules and guidelines for SAP OData V4 objects used as sources and targets in mappings and mapping tasks:

- When you use a simple source filter with a flat source object that contains data of the Date or TimeOfDay data types, ensure that you use the following format in the filter condition:
  `YYYY-MM-DD HH24:MI:SS.MS`
- When you read hierarchical data of the Date or TimeOfDay data types, ensure that you enable the **Optimize transformation** option in the Source transformation.
- When you write hierarchical data to SAP S/4HANA applications, you can only perform an insert operation on the target.
- You can read data only from analytical models that are based on a fact, dimension, and table model. If the analytical model is based on an entity-relationship model, the mapping read operation fails.

CHAPTER 4

Data type reference

Data Integration uses the following data types in mappings, dynamic mapping tasks, and mapping tasks with SAP OData V4.

Native data types

SAP OData V4 data types appear in the Source and Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformation. They are internal data types based on ANSI SQL-92 generic data types, which Data Integration uses to move data across platforms. Transformation data types appear in all remaining transformations in a mapping, dynamic mapping task, or mapping task.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Flat SAP OData V4 and transformation data types

Use only the compatible flat SAP OData V4 data types in mappings, dynamic mapping tasks, and mapping tasks to read from SAP Datasphere and SAP S/4HANA applications, or write to SAP S/4HANA applications.

The following table lists the flat SAP OData V4 data types that you can use with SAP OData V4 Connector, along with the corresponding transformation data types when you read from SAP Datasphere applications:

SAP OData V4 Data Type	Transformation Data Type
Edm.Binary	Binary
Edm.Boolean	String
Edm.Date	Date/Time
Edm.DateTimeOffset	Date/Time
Edm.Decimal	Decimal
Edm.Double	Double
Edm.Int32	Integer

SAP OData V4 Data Type	Transformation Data Type
Edm.Int64	Bigint
Edm.Single	Double
Edm.String	String
Edm.TimeOfDay	Date/Time

The following table lists the flat SAP OData V4 data types that you can use with SAP OData V4 Connector, along with the corresponding transformation data types when you read from or write to SAP S/4HANA applications:

SAP OData V4 Data Type	Transformation Data Type
Binary	Binary
Byte	Integer
Date	Date/Time
Decimal	Decimal
Double	Double
Int16	Decimal
Int32	Integer
Int64	Bigint
String	String
TimeOfDay	Date/Time

Hierarchical SAP OData V4 and transformation data types

Use only the compatible hierarchical SAP OData V4 data types in mappings, dynamic mapping tasks, and mapping tasks to read from SAP S/4HANA applications.

The following table lists the hierarchical SAP OData V4 data types that you can use with SAP OData V4 Connector, along with the corresponding transformation data types when you read from or write to SAP S/4HANA applications:

SAP OData V4 Data Type	Transformation Data Type
Binary	String
Byte	Integer
Date	String
Decimal	Decimal
Double	Double
Int16	Decimal
Int32	Integer
Int64	Bigint
String	String
TimeOfDay	String

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